| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **2** | (a) | | A and C and E identified as correct (2)<br>2 correct gains one mark<br>-1 for each additional tick > 3 ticks | 2 | | | 2 | | |
| | (b) | (i) | $x$ = (cos 25° × 12) **–** 5.44 seen **or** $x$ = 5.44 **or** 6 cos 25° **or** use of sin 25 and Pythagoras<br>**or** argument based on C of G of extendable arm in middle therefore $x$ = 2.00 + 3.44 | | 1 | | 1 | 1 | |
| | | (ii) | C.M.= $2.2 \times 10^4 \times 2.00$ (or $4.4 \times 10^4$ seen) (1)<br>A.C.M.= ($W \times 8.88$ **ecf**) + ($3.0 \times 10^3 \times 3.44$) (1)<br>Correct algebra and $W$ shown = 3793 [N] (1)<br>3767 [N] if approximation of 5.5 used<br>3810 [N] if approximation of 5.4 used | | 3 | | 3 | 3 | |
| | | (iii) | 3800 **ecf** **–** 1700 = 2100 [N] (1)<br>Conversion of mass-force i.e. $m = \frac{F}{g}$ (1)<br>= 214 [kg] (1)<br>Conclusion reasonable and consistent with their figures (1)<br><br>**Alternative:**<br>2 people mass approx 200 kg (1)<br>Conversion of mass-force i.e. $F = mg$ (1)<br>Adding both forces **–** giving around 3700 [N] depending on mass (1)<br>Conclusion reasonable and consistent with their figures **ecf** on comparison with (b)(ii) (1) | | | 4 | 4 | 2 | |
| | | | **Question 2 total** | **2** | **4** | **4** | **10** | **6** | **0** |